# Making the Right Connections to Improve the Comprehension of Struggling Readers

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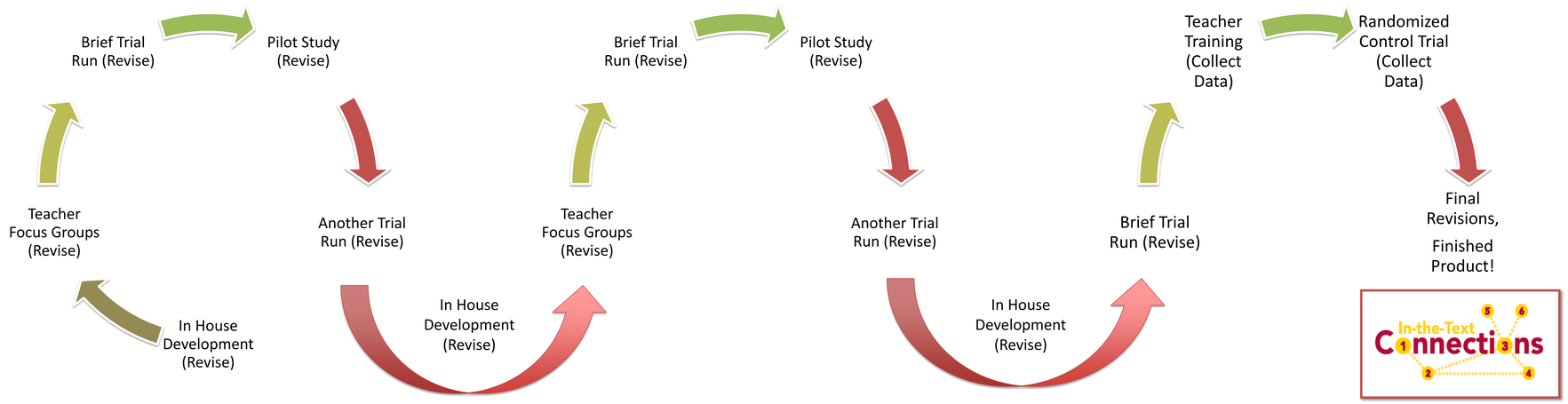
UNIVERSITY OF MINNESOTA
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In-the-Text Connections

Practice | Empirical Evidence | Theory

## BUILDING BLOCKS

## ITERATIVE DEVELOPMENT OF A READING COMPREHENSION INTERVENTION



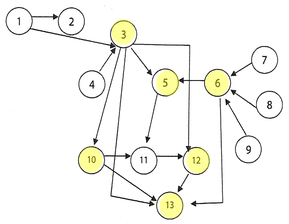

### THEORY: Cognitive Psychology

Cognitive psychologists emphasize the importance of constructing a 'coherent representation of text' (van den Broek, 2009).

To construct a coherent representation of a text, the reader must make connections within text and to background knowledge. Further, the reader must access, integrate, and apply these text- and knowledge-based connections.

A coherent representation of text includes the most important parts of the text—the events or ideas that are most highly connected to each other.

1. Once upon a time, there lived a Princess named Lenore
2. who was ten years old.
3. One day, Princess Lenore became ill
4. from eating too many chocolate-raspberry tarts.
5. The royal physician was called to come
6. to make Lenore feel better.
7. He took her temperature,
8. felt her pulse,
9. and made her stick out her tongue.
10. Knowing that the King was concerned,
11. the physician brought him in after he was finished with the examination.
12. When the king saw his daughter lying there,
13. he asked if there was anything she wanted.




When asked to recall or summarize texts, good readers tend to include the most important, highly connected events, and leave out events with few connections. But poor readers often struggle to do so: They might not distinguish between important events and less important details, and they may even get side-tracked by tangential information in the text.

Our reading intervention, "In-the-Text Connections," involves asking "causal" (Why or How) questions that are designed to help readers focus on the most important, highly-connected parts of the text.

### PRACTICE

We have gathered input and feedback from classroom teachers and specialists, tutors, and students through focus group discussions and questionnaires. This information has informed our efforts to improve the feasibility and appeal of the intervention to be implemented in schools.

### EMPIRICAL EVIDENCE

We have gathered empirical data to determine the promise of the intervention to improve struggling readers' comprehension, including:

- Observations of implementation fidelity and quality
- Student reading data:
  - ✓ Questions answered before and after prompting
  - ✓ Recall of text
  - ✓ Oral reading fluency

## Year 1

**Research Question:**
What *type* and *timing* of questioning is most feasible and promising for struggling readers?

***Theory: Question type and timing matter.***

**Type**

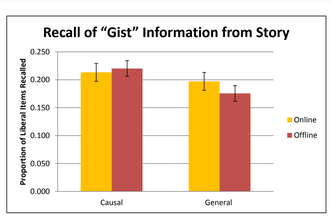
***Causal***:
- Why was the royal physician called?
- Why was the King concerned?

***General***:
- How does the current sentence connect to the rest of the story?

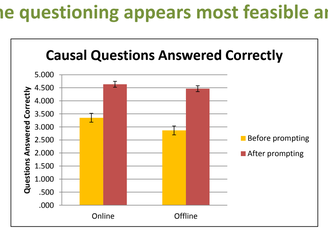
**Timing**
- ***During reading*** (ONLINE, while the reader is constructing a mental representation of text)
- ***After reading*** (OFFLINE, so as not to interrupt the reader's construction of a mental representation of text)

***Practice: Need a structure that is feasible in schools.***
- Small group (3-5 students)
- 3 x per week; 30 min per session, for 6 weeks
- Pre-teaching of key vocabulary (~3-5 min)
- Reading & making connections (~20 min)
- Wrap-Up (~2 min)

***Empirical Evidence:***




Causal Online > General Offline; **ES = .51**, $p < .05$
Causal Offline > General Offline; **ES = .59**, $p < .05$
Tutors found *causal online* questions more feasible to implement.
**Causal online questioning appears most feasible and promising.**




**Students made better connections after prompting.**

## Year 2

**Research Question:**
How can we best *individualize* the intervention for struggling readers?

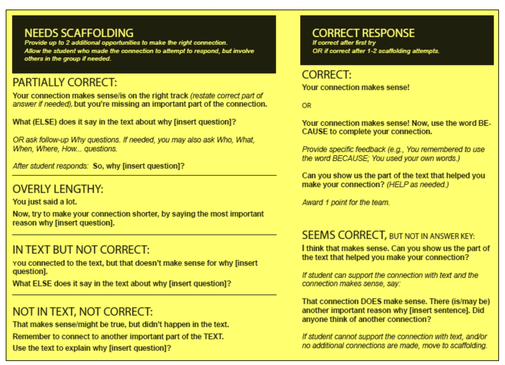
***Theory: Type of feedback matters.***

**Direct Feedback**
- Explicit modeling that shows students how to make a good connection.
- Pro: Efficient (?)
- Con: Less responsive to individual students.

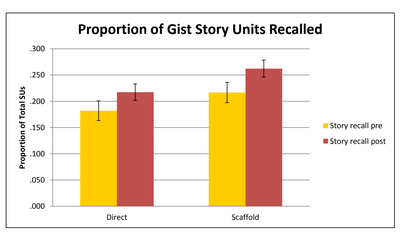
**Scaffolded Feedback**
- Questioning designed to guide student to make a good connection.
- Pro: Responsive to individual students
- Con: Less efficient (?)

***Practice:***

*Scaffolded feedback based on what we have learned about student responses to questions.*




***Empirical Evidence:***

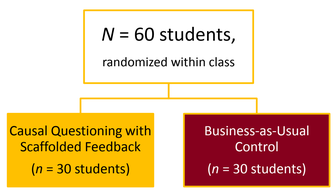



Main effect of time, $p = .008$
No main effect of feedback condition
**Tutors found scaffolded feedback more feasible and desirable.**

## Year 3

**Research Question:**
Is the intervention *feasible* and *promising* when implemented in school settings?

***Setting & Participants***
- Two Twin Cities area districts (urban and suburban)
- 7 schools, 20 classrooms
- 60 fourth-grade struggling readers (bottom 25th percentile in comprehension, but read at least 90 correct words per min in fall)
- 4 groups implemented by UMN tutors; 4 groups implemented by school personnel (classroom teachers or reading specialists)




***Measures***

Proximal:
- Questions answered with and without prompting
- Highly-connected text units recalled
- Important text units highlighted
- Oral reading fluency

Distal:
- Maze (correct word choices)
- Multiple-choice Online Cloze Comprehension Assessment (MOCCA)

***Stay tuned for results!***

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